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# Sound gaps boost emotional audiovisual integration independent of attention: Evidence from an ERP study

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ABSTRACT

The emotion discrimination paradigm was adopted to study the effect of interrupted sound on visual emotional processing under different attentional states. There were two experiments: Experiment 1: judging facial expressions (explicit task), Experiment 2: judging the position of a bar (implicit task). In Experiment 1, ERP results showed that there was a sound gap accelerating the effect of P1 present only under neutral faces. In Experiment 2, the accelerating effect (P1) existed regardless of the emotional condition. Combining two experiments, P1 findings suggest that sound gap enhances bottom-up attention. The N170 and late positive component (LPC) were found to be regulated by emotion face in both experiments, with fear over the neutral. Comparing the two experiments, the explicit task induced a larger LPC than the implicit task. Overall, sound gaps boosted the audiovisual integration by bottom-up attention in early integration, while cognitive expectations led to top-down attention in late stages.

## 1. Introduction

When you watch a movie, you can tell by the protagonist's frightened or surprised voice combined with his shocked face that he has, say, discovered a burglar, and has not walked in on a surprise birthday party. This recognition process involves emotional audiovisual integration, which is the convergence or integration of multisensory signals to synthesize an internal mental representation. This process can maximize the effectiveness of information transmission in natural environments and improve the accuracy of perception (Parise & Ernst, 2016). Thus, emotional audiovisual integration is an effective method for transmitting and decoding emotional information and is of great significance for human survival and evolution.

Currently, researchers understand that in this process, an emotion signal in one modality boosts or modifies the perception of a signal in another modality (Ball, Michels, Thiele, & Noesselt, 2018). According to previous studies, three main possibilities of audiovisual integration can be envisaged based on the emotional dimension. First, when both visual and auditory channels contain consistent emotional information, the cognitive resources required for audiovisual integration are greatly reduced and the perception of emotion is enhanced (Klasen, Chen, & Mathiak, 2012; Pourtois & Dhar, 2013). Second, when only the auditory channel contains emotional information, studies have shown that emotion-encoding sound affects non-emotion visual processing, but the specific mechanism has not been identified (Zeelenberg & Bocanegra, 2010; Harrison & Davies, 2013). Finally, in the case of non-emotive auditory signals integrated with emotion-encoding visual stimuli, from visual to auditory perspective, several studies revealed that emotional visual cues increase the individual's level of auditory attention and improve the coding of neutral sounds (Sugimoto, Nittono, & Hori, 2007; Tartar, de Almeida, McIntosh, Rosselli, & Nash, 2012). From an auditory to visual perspective, the conclusion is still unclear. In the earlier research, Schupp and Stockburger (2008) concluded that emotion processing in the visual domain is not affected by auditory task demands. Pápai and Soto-Faraco (2017) recently used a binocular rivalry paradigm to investigate whether an auditory event consisting of an interruption of a usually continuous sound (a tone) would affect the audiovisual integration effect below the threshold of awareness. In previous studies, researchers have usually promoted audiovisual integration by raising or lowering the pitch of a meaningless sound or presenting a pitch that deviates completely from the original pitch, whereas the interrupted sound is a period of blanking out in the middle of a

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continuous sound (Sugimoto *et al.* 2007). The sound gap in anticipation weakens any sound-flash integration based on simple neuronal summation, as the gap disrupts the continuity of information on the auditory channel. However, the results showed that the use of gapped sound increased the probability of audiovisual integration. This implies that the effects on audiovisual integration of interrupted versus continuous sounds represent different neural mechanisms. Nevertheless, the integration of dual-channel information is influenced by a variety of factors, such as task requirements and attentional status, thus making it difficult to draw a consistent conclusion.

In recent decades, many studies have used the event-related potential (ERP) technique to examine the neural mechanisms (Cecere, Gross, Willis, & Thut, 2017; Kokinous, Kotz, Tavano, & Schröger, 2015) and time course of the emotional multisensory integration (MSI) (Gao, Weber, & Shinkareva, 2019; Pourtois, De Gelder, Vroomen, Rossion, & Crommelinck, 2000). The ERP components P1 (Linkenkaer-Hansen *et al.* 1998), N170 (Bentin, Allison, Puce, Perez, & McCarthy, 1996), and LPC (late positive component) (Rellecke, Sommer, & Schacht, 2012) are suitable for representing the effects of emotional MSI. First, in the early stages of audiovisual integration, a larger P1 was evoked from current sources in the striatal (primary visual) cortex by the presence of emotional prosody compared with that of neutral prosody (Brosch, Grandjean, Sander, & Scherer, 2009). And studies had demonstrated that visual and auditory information can be integrated at an earlier stage (~50 ms after stimulation) (Talsma, Doty, & Woldorff, 2007; Tang, Wang, & Peng, 2021). Second, N170 was the classic component for face sensitivity in the occipito-temporal region (Bentin et al., 1996). It is related to the decoding of facial configuration information alongside being sensitive to the cross-modal consistency of emotion signaling (Stefanou et al., 2019). Third, LPC reflected cognitive emotional processing and the extraction of information from emotional memory (Schacht & Sommer, 2009; Sanquist, Rohrbaugh, Syndulko, & Lindsley, 1980).

According to the parallel integration model, MSI is affected by both early bottom-up (Navarra, Alsius, Soto-Faraco, & Spence, 2010) and late top-down attention mechanisms (Xu, Kolozsvári, Oostenveld, Leppänen, & Hämäläinen, 2019; Meijer, Mertens, Pennartz, Olcese, & Lansink, 2019). At the same time, tasks with different attention levels affect the perception of facial expressions and are reflected in the ERP components (P1, N170, EPN, LPC, etc.) (Schindler & Bublatzky, 2020). For example, the emotional effect on P1 was not significantly affected by the task, regardless of whether the subject was focused on the face or emotion or was distracted (Schindler & Bublatzky, 2020; Acunzo, MacKenzie, & van Rossum, 2019). But, recently, it has been found that the differences in the intensity of expressions and low-level visual differences enhance P1, even when using face-scramble images (Schindler, Bruchmann, Gathmann, Moeck, & Straube, 2019). The N170 component was mostly modulated by emotion in different attentional tasks, especially passive observation tasks (Almeida *et al.* 2016; Schindler et al., 2019). For the LPC component, the emotional effect of LPC occurred when attention was sufficiently allocated to the face, especially when confronted with threatening information (Rellecke et al., 2012). In general, all three components were modulated by attentional states and emotion types. Most studies have focused only on the effects of visual senses, which we considered together with auditory channel information. From this, the attentional state, an important and less explored factor in emotional audiovisual integration studies, deserves to be investigated for this purpose.

Furthermore, the three stages of the neurobiological model of affective audiovisual integration proposed by Brück, Kreifelts, and Wildgruber (2011) provided us with a framework to explain the cognitive processing embedded in the various ERP components. In the feature extraction stage, face features are processed along with the pathway from the primary visual cortex (V1) to the fusiform face gyrus (FFA). Acoustic features are within the primary auditory cortex (A1) and the middle superior temporal cortex and the attentional network plays a crucial role in this stage (Wildgruber, Ethofer, Grandjean, & Kreifelts, 2009; Tseng *et al.* 2015). Then, in the information integration stage, the extracted information is transmitted to the posterior superior temporal cortex for further processing. During the cognitive elaboration stage, the dorsolateral frontal and orbitofrontal cortex evaluate the emotional information collected in the first two stages. This cerebral network model reflects a complex multilevel process and its internal information transmission steps.

The purpose of the present study was to investigate the temporal dynamics of the sound gap in emotional MSI under different attentional states using ERP techniques. In Experiment 1, the facial emotion recognition paradigm was used. We designed it to test whether meaningless sounds could influence the processing of emotional signals, to elucidate the neural principle of interrupted sound affecting emotional MSI, to extend the physiologically based model. To further explore whether there was also a facilitating effect of interrupted sounds under the implicit attentional state, we set up Experiment 2, an implicit emotional task, to control for bottom-up facial emotion processing. The two experiments were intended to verify from the perspective of emotional audiovisual integration that early integration is mostly automatic, whereas late integration is regulated top-down and requires conscious awareness. We used EEG technology, combined with methods of ERP analysis, microstate analysis, and source localization analysis, to answer this question.

First, we measured the P1, N170, and LPC amplitudes and examined their differences across conditions. We hypothesized that interrupted sounds would increase subjects' attention in the early stages, whereas ERP components in the middle and late stages were more influenced by the emotional type of the face and that the late stages were modulated by both task type and emotion. We expected that interrupted sound would enhance the audiovisual integration reflecting a larger P1 amplitude (approximately 50 ms after the sound gap appears) in two experiments regardless of the emotional face. Fearful faces evoked greater N170 and LPC than neutral faces, and explicit tasks evoked greater LPC than implicit tasks.

Second, a microstate analysis was used to identify and compare the topologies of the scalp ERP and track changes in the functional state of the brain (Murray, Brunet, & Michel, 2008; Koenig, Stein, Grieder, & Kottlow, 2014). Microstate analysis can capture any dynamic differences between conditions and quantify them as the statistics of microstate activation intensity. Discovering the dynamics of audiovisual integration is crucial for a better understanding of the neural course of integration and network activation. We hypothesized that in the early, relatively fixed stage, the microstate dynamics of the two tasks would not differ significantly, whereas in the later stages, the dynamics would be task-specific, and different task transitions and task modes would occur.

## 2. Materials and method

### 2.1. Participants

We performed sample size estimation using G*Power 3.1 (Faul, Erdfelder, Buchner, & Lang, 2009). Based on a medium effect size (Cohen's $f = 0.25$; Cohen, 2013), a sample of at least 24 participants were needed ($\alpha = 0.05$, 1-$\beta = 0.80$). In Experiment 1&2, a total of 28 healthy male volunteers between the ages of 29 and 48 years were recruited ($M \pm SD$: 34.14 ± 5.04 years). All the participants reported normal auditory and normal or corrected-to-normal visual acuity and were free of neurological or psychiatric problems. The study was approved by the Ethics Committee of Liaoning Normal University (China), and the investigation was conducted in accordance with the latest version of the Declaration of Helsinki. Before the experiment, all participants signed the informed consent and were paid after the experiment.

### 2.2. Stimuli

There were 10 neutral and fearful facial pictures respectively selected from the native Chinese Facial Affective Picture System (CFAPS), included 10 females and 10 males (Fig. S1). The fearful and neutral pictures differed significantly in valence [$t$ (9) = 21.11, $p$ < 0.001 ($M \pm SE$, Neutral: 5.05 ± 0.10, Fearful: 6.95 ± 0.37)] and arousal [$t$ (9) = 13.16, $p$ < 0.001 ($M \pm SE$, Neutral: 2.79 ± 0.17, Fearful: 6.26 ± 0.73)]. Each facial picture was cropped to 6.8 × 8.6 cm (6.5 × 8.2° of view angle), and other physical properties of the pictures were conformable.

The auditory stimuli in this study consisted of a total of 2 sinusoidal meaningless sounds, each tone was 60 dB at a sampling rate of 44100 Hz. One is a continuous sound, which lasts 1000 ms. The other is an interrupted sound, referring to a continuous sound interrupted by a short silent gap that constitutes the auditory stimulus. The interruption period ranges from 100 ms to 200 ms, and the sound continues to be presented at 201 ms, with the interrupted sound also lasting 1000 ms (Fig. S1).

### 2.3. Procedure and design

Each participant was seated comfortably in a sound-attenuated room. Stimulus presentation was controlled using the E-prime software. In total, the experiments included 4 unisensory conditions (only interrupted sound, only uninterrupted sound, only fear face, only neutral face) + 4 audiovisual conditions (2 sound types×2 emotional faces) for a total of 8 conditions, each condition including 64 trials, for a total of 512 trials. The test phase included eight blocks, each comprising 64 trials. As shown in Fig. 1, each trial began with a fixation cross presented for 500 ~700 ms. Then the target stimulus was presented for 1000 ms which included the audiovisual stimulus or the only visual or only auditory stimulus. Then a white bar would appear randomly on the screen for 350 ms, the position of the bar would be higher or lower than the eye or mouth of the face. In Experiment 1 participants were asked to report the emotion of the target by pressing the "F" or "J" button on the keyboard with both index fingers. The audiovisual stimuli included four types: 2 visual stimuli (neutral, fearful) × 2 auditory stimuli (uninterrupted, interrupted); the only visual stimuli consisted of two types: neutral and fearful face and the only auditory stimulus included uninterrupted and interrupted sounds. All stimuli were randomized and counterbalanced across participants. In Experiment 1, we asked subjects to make emotional judgments based on the information they heard and saw. The procedure of Experiment 2 was identical to that of Experiment 1, with the only difference being that the tasks to be completed by the participants were different. In Experiment 2, the participants were needed to judge the position of the white bar, which was higher or lower than the eye or mouth in a total of four conditions (higher than eye, higher than mouth, lower than eye, lower than mouth). When the bar is above the eye or the mouth, the uniform response is higher, and when the bar is below the eye or the mouth, the uniform response is lower.

### 2.4. EEG recordings and ERPs data analysis

EEG data were recorded by a 64-channel EEG recording system of Brain Products ERP (Germany). EEG signals were recorded with a 0.01–100 Hz bandpass filter and a continuously sampling rate at 500 Hz/channel. The online reference was left mastoid and the impedances of the electrode point were below 5 kΩ.

ERP data were preprocessed using EEGLAB, an open-source toolbox running under the MATLAB environment (Delorme & Makeig, 2004). The raw data were first high pass filtered at 0.1 Hz and low pass filtered at 30 Hz and adjusted the offline re-reference to REST (Dong et al., 2017). The data with large artifacts (exceeding ± 75 μV) were removed and channels with poor signal quality were interpolated. Using the independent component analysis algorithm (Delorme & Makeig, 2004), trials contaminated by eye blinks and other artifacts were detected and removed. The data were segmented to 1000 ms epochs time-locked to stimuli onset, starting 200 ms prior to stimuli onset. The epochs data were baseline corrected using the 200 ms before the stimuli onset. More than 80% of the trials were retained in further analysis. ERP waveforms were computed separately according to conditions for audiovisual conditions. For the unisensory conditions, we summed two by two according to the type of audiovisual conditions that occurred to obtain the audio+visual (A+V) conditions (only interrupted sound+only fear face, only uninterrupted sound+only neutral face, only interrupted sound+only fear face, only uninterrupted sound+only neutral face).

First, the two experiments performed the statistical analysis of ERP components separately, and the following parameters were used in both experiments. The average amplitude of the P1 component (124–164 ms) and N170 component (196–236 ms) was measured at posterior electrode-sites (P7/PO7/P8/PO8). For LPC, we selected a time window

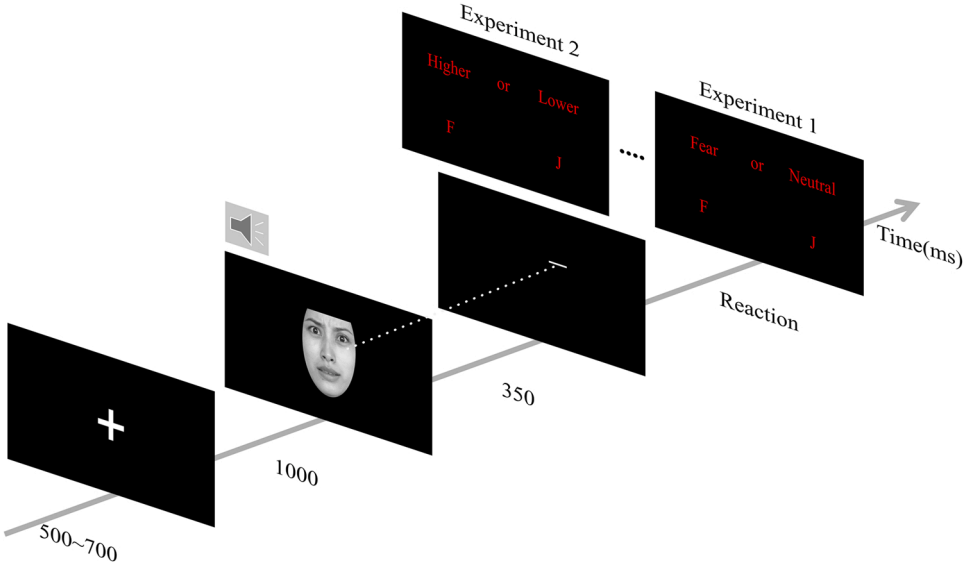



**Fig. 1.** The figure shows the experimental procedure of a single trial in Experiment 1. In Experiment 1, participants were needed to judge whether the facial expression is neutral or fearful. The only difference between the two experiments was that the judgment task. In Experiment 2, participants judged whether the bar appeared above or below the eyes or mouth.

of 300–800 ms and selected the CPz, Pz, CP1, CP2, P1, and P2 electrodesites. The ERP data were subjected to a three-way repeated-measures analysis of variance (ANOVA) with a 2 (Modality: audiovisual and audio + visual) × 2 (Emotion: Neural and Fear) × 2 (Auditory: Uninterrupted and Interrupted) design. Finally, we performed joint statistics on the three components of the two experiments. We performed a four-way repeated-measures analysis of variance (ANOVA) with a 2 (task: Explicit and Implicit) × 2 (Emotion: Neural and Fear) × 2 (Auditory: Uninterrupted and Interrupted) × 2 (Modality: audiovisual and audio + visual) design. For the analysis, the $p$ values were corrected by Greenhouse–Geisser correction.

After the ERPs analysis, we performed Topographic Consistency Test (TCT) analysis (details in the appendix) and microstate analysis on Randomization Graphical User interface (RAGU) (Habermann, Weusmann, Stein, & Koenig, 2018). The identification of microstate prototype maps was based on the K-mean algorithm. The optimal number of microstates was jointly determined by the global explained variance and the cross-validation criterion (Koenig et al., 2014). Then we fit the obtained microstate prototypes back to the grand averaged data of the four conditions of the two experiments (Lehmann, Ozaki, & Pal, 1987). Statistical analysis of microstate parameters was based on randomization statistics. Randomization statistics compared differences of particular microstate features between real data sets against the distribution under the null hypothesis (Koenig & Melie-García, 2010).

## 3. Results

### 3.1. Behavioral results

In Experiment 1 for RT ($M \pm SE$), the main effect of interruption was significant [$F$ (1, 27) = 4.367 $p$ = 0.046, $\eta_p^2$ = 0.139]. This means that under interrupted sound condition (439.41 ± 16.29 ms) the reaction was faster than under uninterrupted sound (448.84 ± 16.62 ms). The main effect of emotion [$F$ (1, 27) = 0.303, $p$ = 0.587, $\eta_p^2$ = 0.011] and interaction effect between Interruption × Emotion [$F$ (1, 87) = 0.014, $p$ = 0.905, $\eta_p^2$ = 0.001] were not significant (Fig. 2).

For accuracy, only the significant main effect of Emotion was observed [$F$ (1, 27) = 8.569, $p$ = 0.007, $\eta_p^2$ = 0.241] It indicated that the accuracy was higher in neutral condition (98.90 ± 0.4%) than in fear condition (93.90 ± 1.7%). The main effect of interruption [$F$ (1, 27) = 2.579, $p$ = 0.120, $\eta_p^2$ = 0.087] and interaction effect between Interruption × Emotion [$F$ (1, 27) = 0.002, $p$ = 0.966, $\eta_p^2$ = 0.000] were not significant (Fig. 2).

In Experiment 2 for RT, the main effect of interruption [$F$ (1, 27) = 0.424, $p$ = 0.520, $\eta_p^2$ = 0.015], the main effect of emotion [$F$ (1, 27) = 0.173, $p$ = 0.681, $\eta_p^2$ = 0.006] and the interaction between the interruption × emotion [$F$ (1, 27) = 2.609, $p$ = 0.118, $\eta_p^2$ = 0.088] were not significant.

For accuracy, the main effect of interruption [$F$ (1, 27) = 1.734, $p$ = 0.199, $\eta_p^2$ = 0.060], the main effect of emotion [$F$ (1, 27) = 3.053, $p$ = 0.092, $\eta_p^2$ = 0.102] and the interaction between the interruption × emotion [$F$ (1, 27) = 0.518, $p$ = 0.478, $\eta_p^2$ = 0.019] were not significant.

### 3.2. ERPs results

#### 3.2.1. Experiment 1

P1 component. We did not find that the main effect of the three factors was significant. There was a significant three-way interaction effect between Interruption × Emotion × Modality [$F$ (1, 27) = 6.193, $p$ = 0.019 $\eta_p^2$ = 0.187]. We used a post hoc Bonferroni to explore this interaction further and found that when neutral emotion was presented in audiovisual mode, P1 amplitude ($M \pm SE$) was larger in the interrupted condition (4.759 ± 0.428 μV) compared to the uninterrupted condition (4.365 ± 0.374 μV) (Fig. 3). We found no such difference in neutral emotion presented in the audio + visual mode.

N170 component. A significant main effect of emotion was observed [$F$ (1, 27) = 15.213, $p$ = 0.01, $\eta_p^2$ = 0.360], with larger amplitude in fear condition (−4.192 ± 0.482 μV) relative to neutral condition (−3.819 ± 0.450 μV). Another significant main effect of modality was also observed [$F$ (1, 27) = 25.219, $p$ < 0.001, $\eta_p^2$ = 0.483], with larger amplitude in audio + visual condition (−4.444 ± 0.507 μV) relative to audiovisual condition (−3.567 ± 0.434 μV). And we did not find that any interaction effect was significant (Fig. 3).

LPC component. Only a significant main effect of emotion was observed [$F$ (1, 27) = 45.211, $p$ < 0.001, $\eta_p^2$ = 0.626], with larger amplitude in fear condition (1.675 ± 0.220 μV) relative to neutral condition (2.199 ± 0.220 μV).

#### 3.2.2. Experiment 2

P1 component. Only a significant two-way interaction effect between Interruption × Modality was found [$F$ (1, 27) = 7.918, $p$ = 0.009, $\eta_p^2$ = 0.227]. A post hoc Bonferroni on this interaction revealed that under the audiovisual mode, the interrupted condition (4.349 ± 0.310 μV) amplitude was larger than the uninterrupted condition (4.097 ± 0.303 μV), but under audio + visual mode, this difference was not found (Fig. 4).

N170 component. A significant main effect of emotion was observed [$F$ (1, 27) = 5.530, $p$ = 0.026, $\eta_p^2$ = 0.170], with larger amplitude in fear

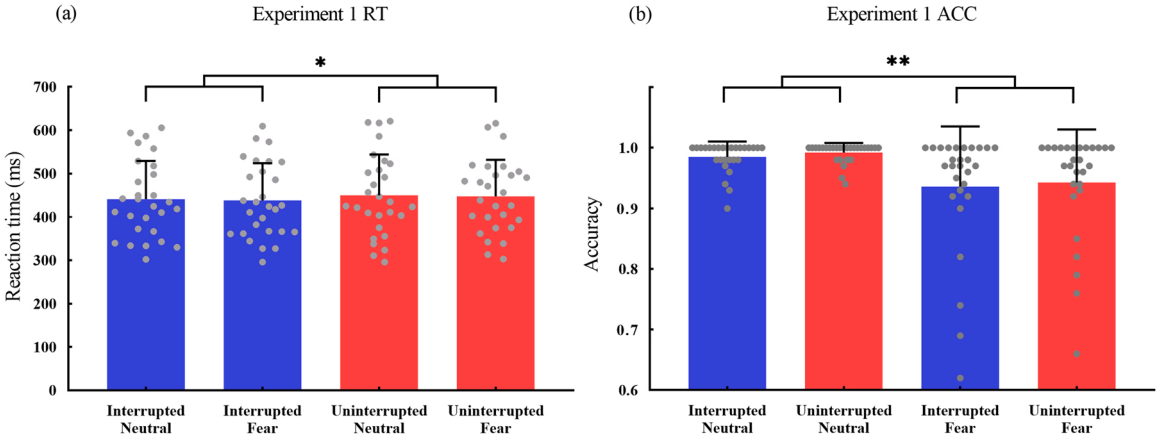



**Fig. 2.** Experiment 1 behavior result histogram (a) Under the interrupted condition, participants had shorter average response times. (b) Under the neutral condition, participants had a higher accuracy.

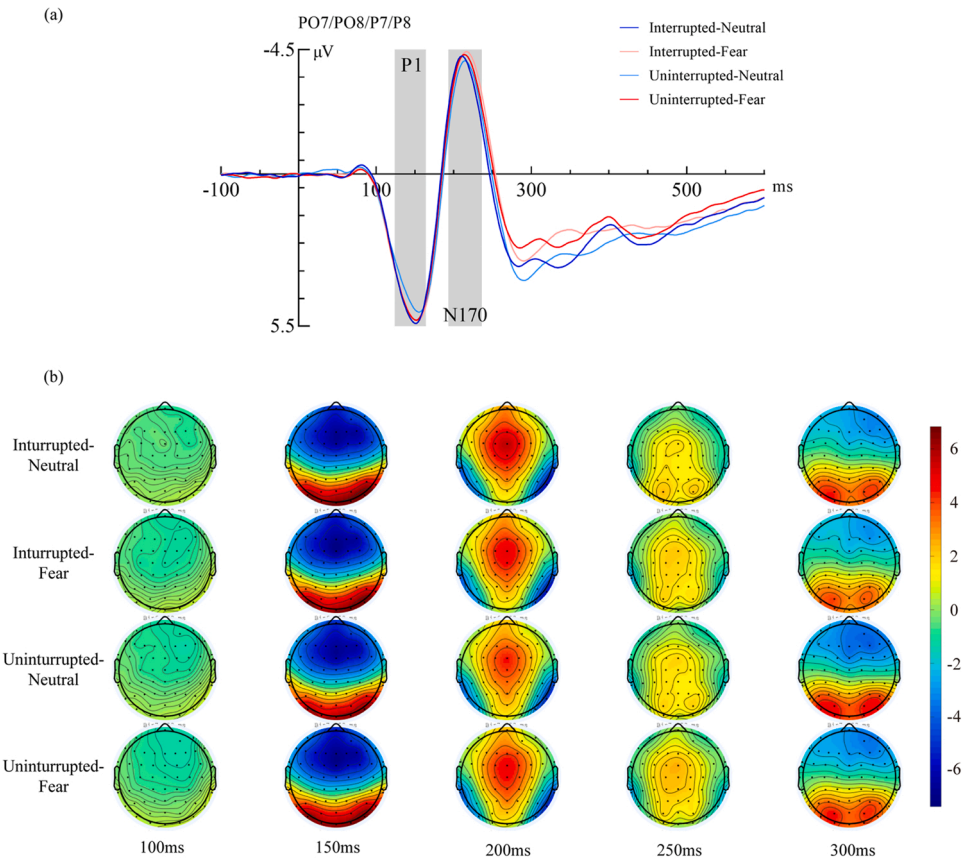



**Fig. 3.** Experiment 1 ERP result for P1 and N170 in 4 conditions. (a) P7, P8, PO7, and PO8 were selected for P1 and N170 and presented together after averaging. In P1, under audiovisual conditions, the interrupted-neutral amplitude was greater than the uninterrupted-neutral. In N170, fearful faces induced greater amplitude. (b) The topographic maps for 4 conditions in Experiment 1 began at 100 ms and ended at 300 ms, with a 50 ms interval between each topographic map.

condition ($-3.236 \pm 0.365$ μV) relative to neutral condition ($-3.015 \pm 0.362$ μV). Another significant main effect of modality was also observed [$F$ (1, 27) = 16.202, $p < 0.001$, $\eta_p^2 = 0.375$] with larger amplitude in audio+visual condition ($-3.391 \pm 0.288$ μV) relative to audiovisual condition ($-2.860 \pm 0.344$ μV). We did not find that any interaction effect was significant (Fig. 4).

LPC component. Only a significant main effect of emotion was observed [$F$ (1, 27) = 10.663, $p = 0.003$, $\eta_p^2 = 0.283$], with larger amplitude in fear condition ($1.016 \pm 0.169$ μV) relative to neutral condition ($0.705 \pm 0.143$ μV).

### 3.3. Comparison between Experiment 1 and Experiment 2

#### 3.3.1. Behavioral result

For the comparison between two experiments, we performed a four-way repeated-measures analysis of variance (ANOVA) with a 2 (Task: Explicit and Implicit) × 2 (Emotion: Neutral and Fear) × 2 (Auditory: Uninterrupted and Interrupted) × 2 (Modality: audiovisual and audio + visual). In comparing the responses time of the two experiments, we only found a task main effect [$F$ (1, 27) = 11.394, $p = 0.002$, $\eta_p^2 = 0.297$], which implies that the implicit task had a longer response time than the explicit task ($M \pm SE$, Explicit task: $444.12 \pm 16.30$ ms; Implicit task: $525.47 \pm 16.75$ ms). We did not find any significant main effects or interactions.

For accuracy, we found a task main effect [$F$ (1, 27) = 44.516, $p < 0.001$, $\eta_p^2 = 0.622$], which implies that the implicit task has a lower accuracy than the explicit task ($M \pm SE$, Explicit task: $0.964 \pm 0.009$; Implicit task: $0.820 \pm 0.024$). We also found an interaction between emotion and task [$F$ (1, 27) = 10.009, $p = 0.004$, $\eta_p^2 = 0.270$], with simple effects analyses showing that only the two emotions differed in accuracy in the explicit task ($p = 0.007$), with neutral faces having higher accuracy than fearful faces (Neutral: $98.90 \pm 0.4\%$; Fear: $93.90 \pm 1.7\%$), while there was no difference between the two emotions in the implicit task ($p = 0.092$). We did not find any other significant results.

#### 3.3.2. ERPs results

P1 component. A significant two-way interaction effect between Interruption × Modality [$F$ (1, 27) = 7.330, $p = 0.021$, $\eta_p^2 = 0.214$] was found. We used a post hoc Bonferroni to explore this interaction further and found that the P1 amplitude in the interrupted condition ($4.542 \pm 0.329$μV) was greater than in the uninterrupted condition ($4.324 \pm 0.300$μV) when subjects were in the audiovisual integration rather than the audio+visual condition.

N170 component. A significant main effect of emotion was found [$F$ (1, 27) = 14.968, $p = 0.001$, $\eta_p^2 = 0.357$], with larger amplitude in fear condition ($-3.714 \pm 0.398$ μV) relative to neutral condition ($-3.417 \pm 0.377$ μV). A significant main effect of task was found [$F$ (1, 27) = 8.123, $p = 0.008$, $\eta_p^2 = 0.231$], with larger amplitude in explicit task condition ($-4.005 \pm 0.464$ μV) relative to implicit task condition ($-3.125 \pm 0.360$ μV). A significant main effect of modality was found [$F$ (1, 27) = 33.215, $p < 0.001$, $\eta_p^2 = 0.552$], with larger amplitude in audio+visual condition ($-3.918 \pm 0.413$ μV) relative to audiovisual condition ($-3.213 \pm 0.366$ μV). We did not find any significant interaction effect in N170.

LPC component. A significant main effect of emotion was observed [$F$ (1, 27) = 47.301, $p < 0.001$, $\eta_p^2 = 0.637$], with larger amplitude in fear condition ($1.608 \pm 0.158$ μV) relative to neutral condition

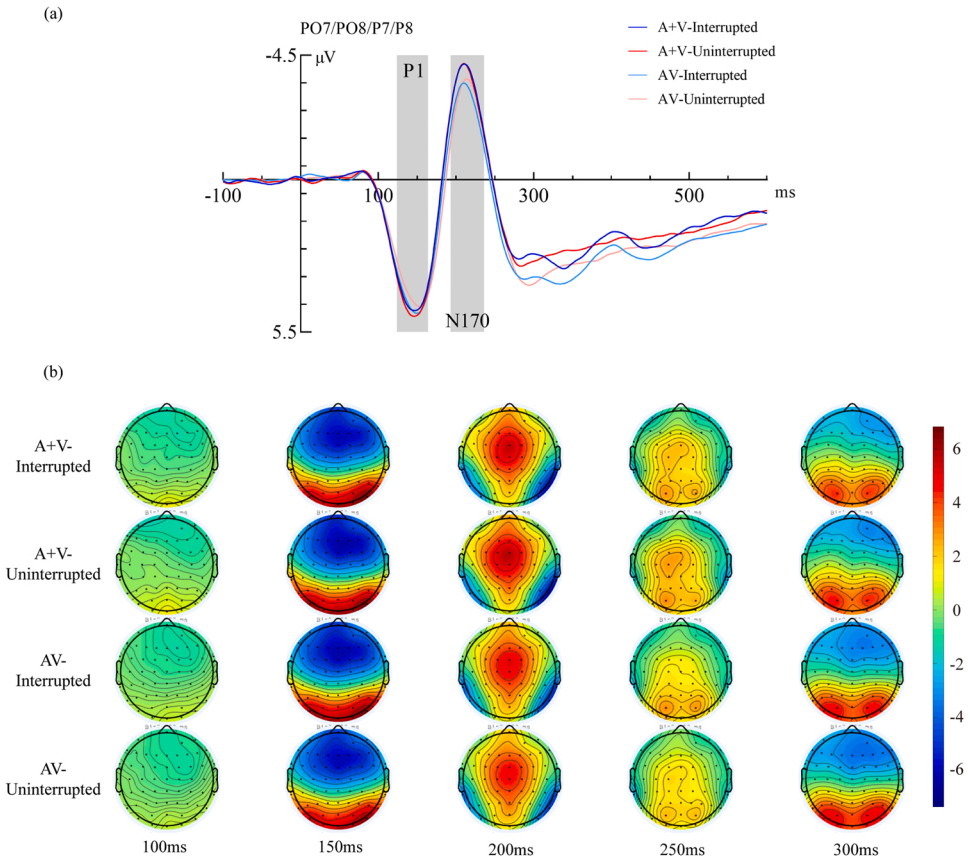



**Fig. 4.** Experiment 2 ERP result for P1 and N170 in 4 conditions. (a) P7, P8, PO7, and PO8 were selected for P1 and N170 and presented together after averaging. In P1, under the audiovisual mode, the interrupted condition amplitude was larger than the uninterrupted condition. In N170 audio+visual condition was larger than the audiovisual condition. (b) The topographic maps for 4 conditions in Experiment 2 began at 100 ms and ended at 300 ms, with a 50 ms interval between each topographic map.

(1.190 ± 0.148 μV). Another significant main effect of task was also observed [$F$ (1, 27) = 24.129, $p < 0.001$, $\eta_p^2 = 0.472$], with larger amplitude in explicit task condition (1.937 ± 0.216 μV) relative to implicit task condition (0.860 ± 0.149 μV). We did not find that any interaction effect was significant (Fig. 5).

### 3.3.3. Microstate result

In Experiment 1 microstate analysis was performed over four conditions. The cross-validation produces identified an optimum of six microstate maps (the test set explained variance over 85%) that was used for subsequent analyses. The fitting of the microstate maps to the grand average ERP between 0 and 900 ms resulted in the same six microstate maps and the same temporal sequence in four experimental conditions. Their respective topographies and the chain of processes are illustrated in Fig. 6. Table 1 shows the details of the onset, offset, and duration of each microstate map in four conditions as well as the significance of the differences obtained from the randomization statistics on microstate map parameters. Comparing the two experiments, before 400 ms, both groups of subjects were performing similar cognitive processing steps. In the later stage, Experiment 1 was mainly undergoing microstate 5 activities, while in Experiment 2, subjects were mostly engaged in microstate 1 activities (Fig. 6). You can see more information in the appendix.

## 4. Discussion

### 4.1. Experiment 1

The three main findings were as follows. P1 component showed that interrupted sound can improve attention allocation in the early stage and only affects audiovisual integration with neutral faces. N170 showed that meaningless sounds had a promoting effect on audiovisual integration, but the interrupted sound could not affect this critical process. The microstate analysis and LPC results showed that in the late stage of advanced cognitive processing, the attentional advantage conferred by the interrupted sound could be consistently maintained, with the result that the emotion discrimination task was completed faster. Taken together, our results show that the presentation of an interrupted sound significantly improves attention distribution while enhancing audiovisual integration and emotion discrimination.

Overall, the P1 results suggest that sound gap not only improves selective attention allocation but also enhances audiovisual integration to enhance the speed of perception of facial emotion (Brosch et al., 2009). Interaction of sound and emotion was only found in the audiovisual condition, which confirmed that the meaningless sounds were involved in facial emotion perception and affected audiovisual integration. Focusing on the interaction effect, we expected that under interrupted sound conditions, P1 should be enhanced in two emotions (Smith, Cacioppo, Larsen, & Chartrand, 2003), but as the result, the interrupted sound produced a larger P1 than did the uninterrupted sound only in the neutral condition. The presence of fear information in the visual channel had a stronger attention-capture effect and a higher level of processing priority (Luo, Feng, He, Wang, & Luo, 2010), thus overwhelming the effect of interrupted sound. Alternatively, fear information has higher attentional capture privileges relative to the continuity of information flow. Obviously, the enhancement effect of the interrupted sound was limited by the visual content. In general, the interrupted sound could participate in visual information processing, but the gain effect of interrupted sound depended on the content of the visual channel.

The most important result of the N170 analysis was that meaningless sounds also contributed to the processing of facial emotion. We found that in the multisensory condition, the N170 was significantly smaller than in the audio+visual condition, showing facilitated under-additivity of MSI. Audiovisual integration also occurs when meaningless sounds

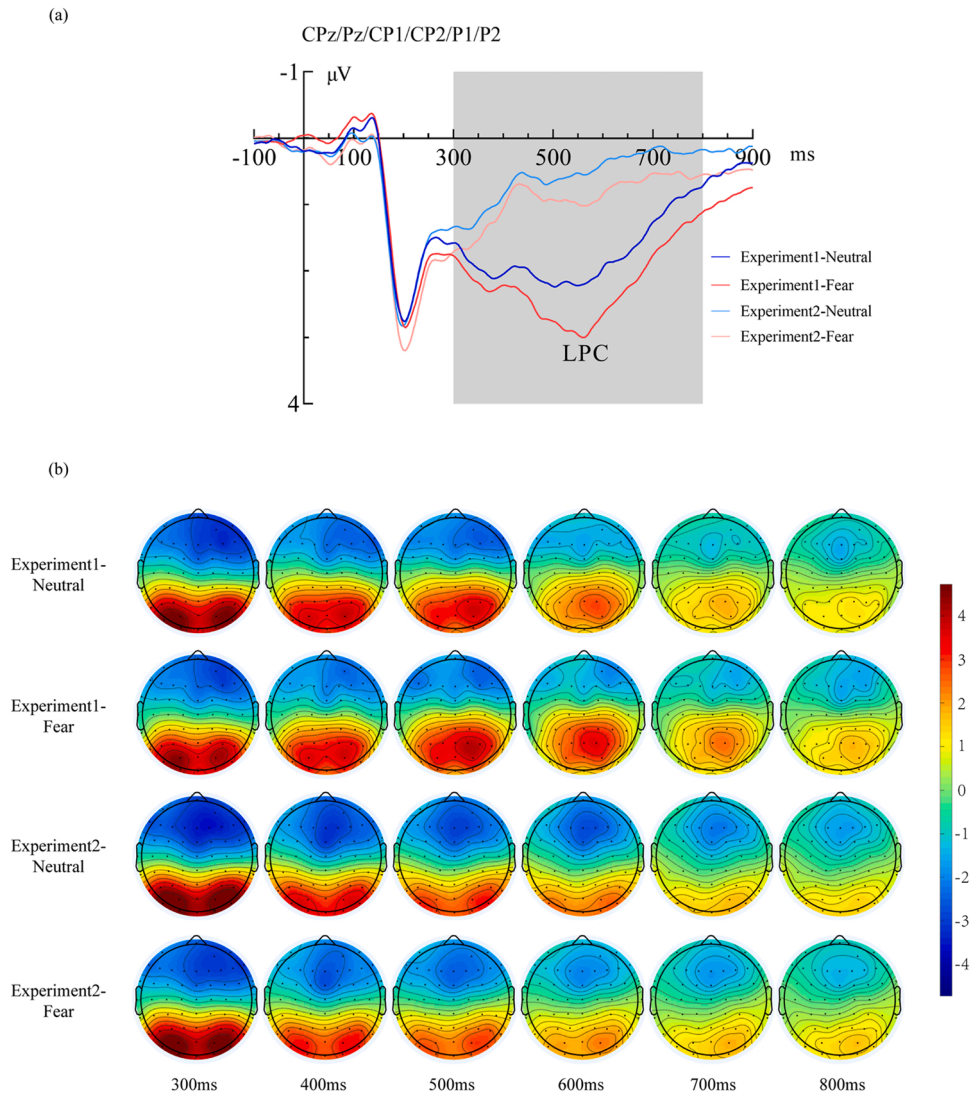



**Fig. 5.** Experiment 1&2 LPC. Experiment 1&2 grand averaged waveforms for LPC in 2 (neutral, fear) conditions. (a) CPz, Pz, P2, CP2, P1, and CP1 were selected for LPC and are averaged over the six electrodes in the figure. Experiment 1 induced a larger LPC than Experiment 2, and the fearful face also induced a larger amplitude. (b) The topographic maps of Experiment 1 and Experiment 2 under fear and neutral conditions ranged from 300 ms to 800 ms with 100 ms intervals.

are presented simultaneously with faces. The brain automatically integrates information presented at the same time, and by default assumes that it always makes sense, or always means something, when information is entered on both channels at the same time (McDonald, Teder-SaÈlejaÈrvi, & Hillyard, 2000). Furthermore, fearful emotion induced a larger N170 amplitude. Fearful expressions anticipate death, attack, or injury, which may be attributed to the significance of fear information for human survival and evolution (Neuberg, Kenrick, & Schaller, 2011). Therefore, the brain automatically and preferentially allocates more attentional resources to the fearful face (Schmidt, Belopolsky, & Theeuwes, 2015).

### *4.2. Experiment 2*

In the implicit task, P1 was not modulated by emotions. The interrupted sound increased the amplitude of P1, implying an increase in the attention levels under interrupted sound. The P1 results suggested that attention is not entirely under bottom-up control and that the "priority" of attention captured by the sound gap increased when emotions were ignored (Pápai & Soto-Faraco, 2017). In other words, even in the early stages, the brain can actively turn off its ability to capture fearful emotions. The results of the N170 analysis were the same as those of Experiment 1. First, the under-additivity of the N170 found during this implicit processing suggested that the mutual activation effect of cross-modal stimuli was not controlled by the task. Instead, it was carried out automatically and spontaneously. Furthermore, the N170 was modulated by the emotion of fear, regardless of the attentional state, a highly automated process that is not subject to the will of the experimental task or the brain.

## 2. General discussion

The two experiments revealed the role of interrupted sound in emotional audiovisual integration and the modulatory effects of the emotion of fear on different ERP components under different attentional states. From the perspective of electrophysiology, we improvised the cross-modal emotional integration model by clarifying the integration method and the influence of attention on the integration process. Thus, we discuss three main issues: the role of meaningless sounds, Attentional states and facial emotions modulating ERP components, and the cross-modal emotional integration model.

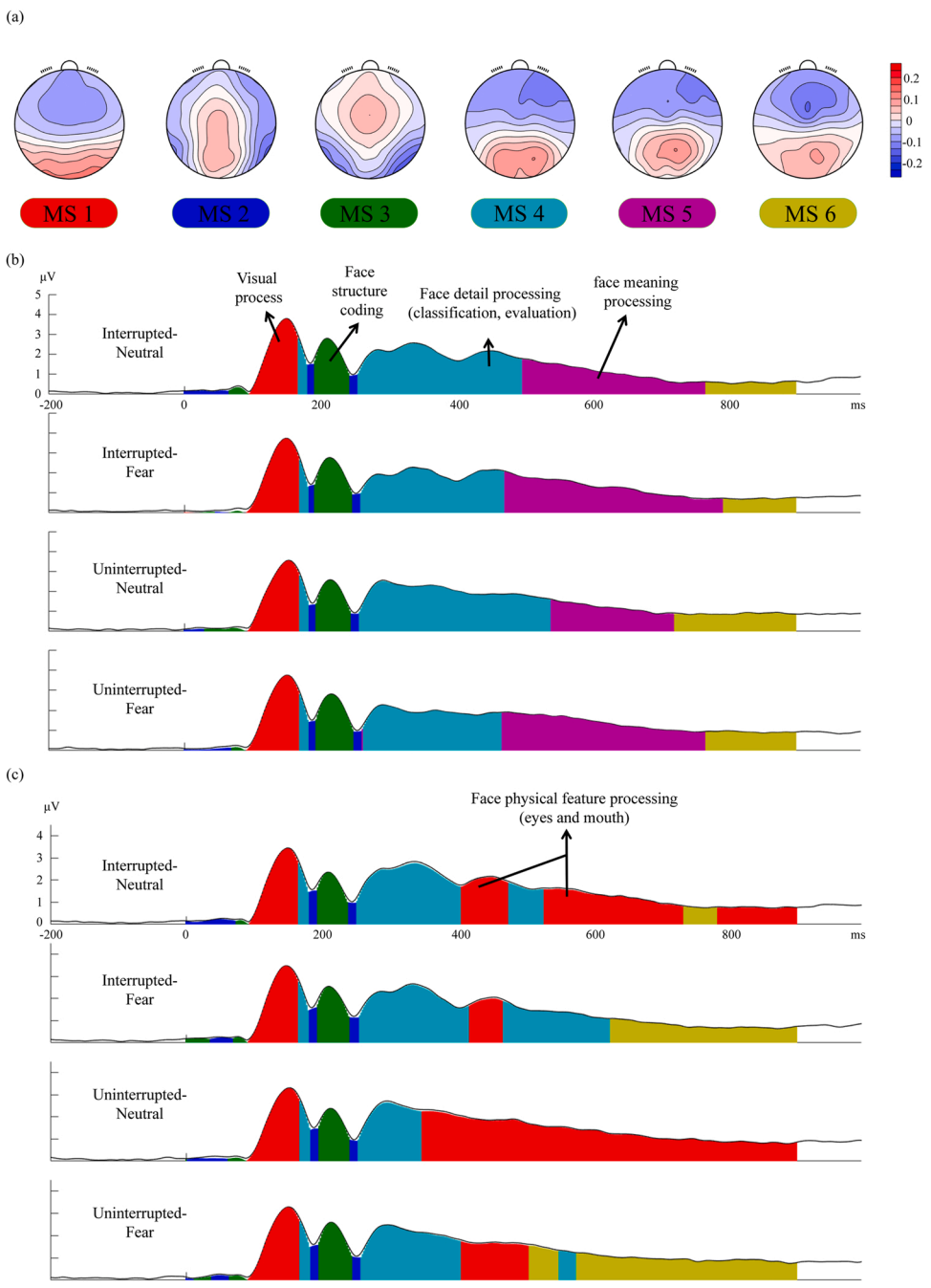



**Fig. 6.** Microstate result. (a) The optimal 6 microstate maps. Blue map areas indicate negative, red areas positive values. All maps have been scaled to have a GFP of one. Different label colors are used to distinguish the microstate map and correspond to the colors in (b) and (c). (MS = microstate) (b) The six-class solution of the microstate assignment for the 4 conditions in Experiment 1. The *y*-axis represents the GFP value (μV) and the *x*-axis represents time (ms). Different colors represent specific microstate classes. Following the time course, MS1 represents the initial processing of visual information, then MS3 means that the brain structurally encodes the face, followed by more detailed information processing of the face in MS4 and finally in MS5 evaluates the meaning of the face and responds. (c) The results of the microstate fitting data of Experiment 2, the interpretation of each index are consistent with the (b) figure. In Experiment 2, the meanings of the first three microstates are consistent with Experiment 1. But at the end of the experiment, participants continued to process the face physical features according to the task requirements.

### *2.1. The role of meaningless sounds*

In the two experiments, we adjusted the attentional mode during the audiovisual integration of emotional faces with meaningless sounds. Combining the two sets of experimental results, we conclude that the interrupted sound caused a larger P1 amplitude, which meant that the interrupted sound attracted more attention than the continuous sound (Smith et al., 2003). Although the interrupted sound may have reduced the simultaneous presentation time of the raw stimulation energy in the two modalities, the information continuity of the auditory modality was interrupted, catching the participants' attention. This was similar to previous behavioral results. Notably, in P1, the boosting effect of interrupted sound is conditional, which drowns when people pay attention to the fear face. Due to this "innate" stimulus advantage (visual dominance, fear capturing attention, etc.), the effect of interrupted sound in enhancing the redistribution of attention was lost (Van Strien, Eijlers, Franken, & Huijding, 2014). Hence, discussing the role of meaningless sounds in emotional picture processing from the perspective of the complexity of the auditory task is the only aspect of this problem and cannot account for the fact that meaningless sounds do not affect the processing of emotional pictures (Schupp *et al.* 2008).

Further, the facilitation effect of interrupted sounds can extend to the stage of face meaning assessment. The microstate results from Experiment 1 showed that microstate 4, representing face detail processing, lasted shorter in the interrupted sound condition, implying that the subjects were able to complete the face detail assessment phase faster (Valdés-Conroy, Aguado, Fernández-Cahill, Romero-Ferreiro, & Diéguez-Risco, 2014). We performed a source analysis of microstate 4 for both experiments and the results showed that the right insula and right superior temporal gyrus had stronger neural responses in the

**Table 1**
Descriptive onset, offset, and duration (in ms) of microstate maps in the 4 conditions (analysis time window 0 ~ 900 ms).

| MS Class | Fitting parameter | Conditions | | | |
|---|---|---|---|---|---|
| | | Interrupted-neutral | Interrupted-fear | Uninterrupted-neutral | Uninterrupted-fear |
| MS 1 RED | Onset(ms) | 98 | 92 | 94 | 94 |
| | Offset(ms) | 168 | 170 | 170 | 170 |
| | Duration(ms) | 72 | 98 | 78 | 78 |
| MS 2 BLUE | Onset(ms) | 182 | 184 | 184 | 184 |
| | Offset(ms) | 256 | 260 | 258 | 262 |
| | Duration(ms) | 22 | 20 | 22 | 22 |
| MS 3 GREEN | Onset(ms) | 194 | 194 | 196 | 196 |
| | Offset(ms) | 244 | 248 | 246 | 250 |
| | Duration(ms) | 52 | 56 | 52 | 56 |
| MS 4 CYAN | Onset(ms) | 170 | 172 | 172 | 172 |
| | Offset(ms) | 498 | 472 | 540 | 468 |
| | Duration(ms) | 256 | 226 | 296 | 218 |
| MS 5 PURPLE | Onset(ms) | 500 | 474 | 542 | 466 |
| | Offset(ms) | 768 | 794 | 722 | 768 |
| | Duration(ms) | 270 | 322 | 182 | 302 |

interrupted condition (Fig. 7) (McCormick, Lacey, Stilla, Nygaard, & Sathian, 2018). This also fits the behavioral results, with shorter responses in the interrupting condition. Combining the above results we speculate that, in the later stage, interrupted meaningless sounds enhanced audiovisual integration efficiency by increasing the speed of information exchange between visual and auditory channels. In summary, the interrupted sound played two main roles: improving attention allocation and enhancing the efficiency of audiovisual integration.

### 2.2. Attentional states and facial emotions modulating ERP components

In the study of facial expressions, ERP was closely linked to various emotions and attentional states, especially fear faces. Despite the controversies in existing findings, we present our perspective on these controversies in light of our study results. On the one hand, the P1 results of studies using explicit attention tasks focusing on emotional faces were inconsistent. Most studies resulted in no emotion regulation effect for P1 with no difference in base emotions compared to a neutral emotion (Schindler & Bublatzky, 2020); however, a small number of

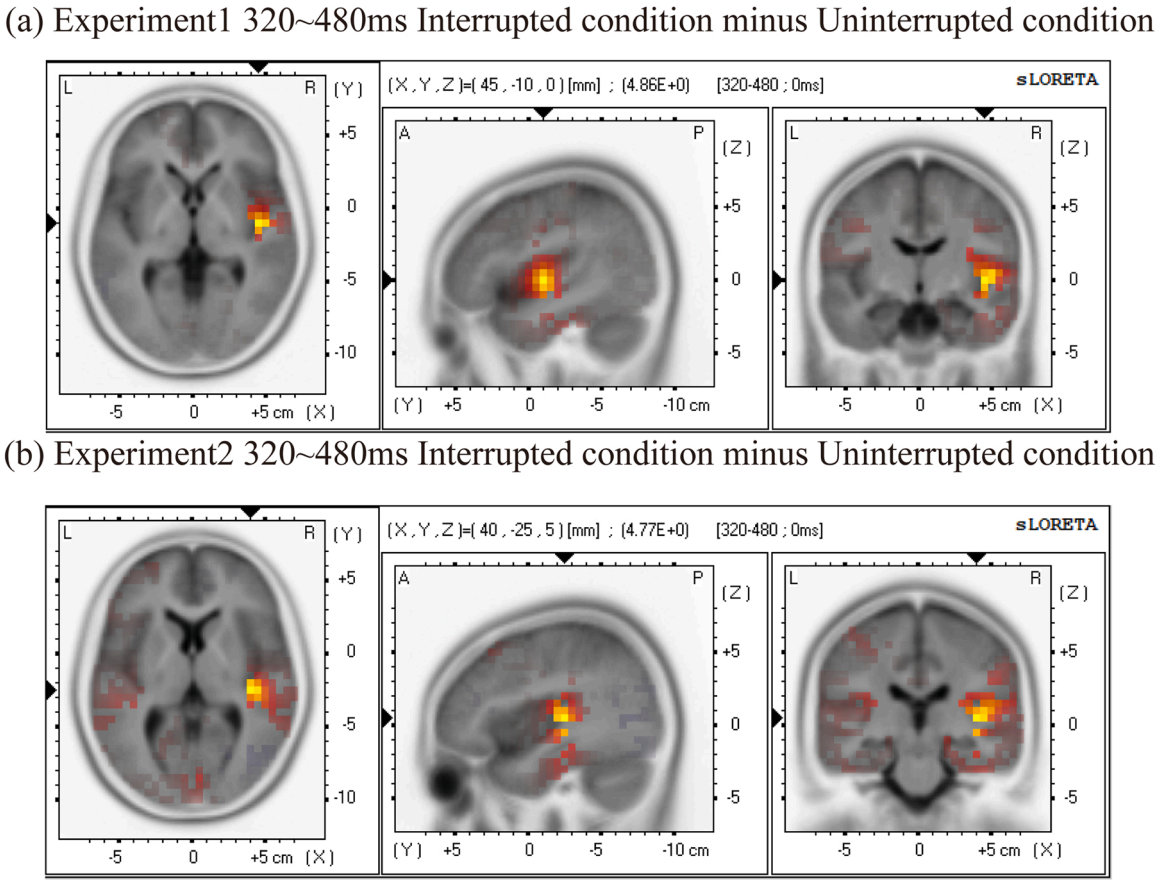



**Fig. 7.** (a) Experiment 1. The right superior temporal gyrus (rSTG) and right insula had stronger neural responses in the interrupted condition than in the uninterrupted condition [$t(0.01) = 4.238$, Extreme $p = 0.0026$]. (b) Experiment 2. The rSTG, bilateral insula, and transverse temporal gyrus had stronger neural responses in the interrupted condition than in the uninterrupted condition [$t(0.01) = 4.296$, Extreme $p = 0.0022$].
Source analysis results of Experiment 1&2.

results yielded the opposite results (Aguado *et al.* 2012; Rellecke et al., 2012). On the other hand, the results of studies with passively observed faces were two-sided, with one part of the results that found no difference between fear and the neutral emotion (Acunzo et al., 2019; Schindler et al., 2019), while others suggested a P1 modulating effect of fear and anger emotions (Blechert, Sheppes, Di Tella, Williams, & Gross, 2012; Smith, Weinberg, Moran, & Hajcak, 2013). The result of Experiment 1 found that P1 was enhanced under the fear condition accompanied by the uninterrupted sound. Interrupted sound enhanced the attentional state in the neutral face condition, and there was no difference in P1 between the fear and neutral faces. Experiment 2 did not find the emotional effect; we only found a P1 modulatory effect for interrupted sounds in the implicit task. Summarizing the results of the two experiments, we argue that the early attentional state or level was influenced by multiple factors (auditory channels, face-specific features, etc.) as different properties all jointly affected P1, leading to inconsistent results. We can confirm that P1 is not a component specific to emotions, but rather an indicator of the attentional state (Smith et al., 2003).

The N170 and LPC results were much more consistent than the P1 results, yielding consistent emotion regulation results for both the explicit attentional state task and the passive viewing of emotional faces, with N170 or LPC waves significantly larger in the fear condition than in the neutral condition (Almeida et al., 2016; Schindler et al., 2019; Schindler & Bublatzky, 2020). Results from the two experiments revealed the main emotional effects of both the N170 and LPC. By incorporating microstate analysis, a more elaborate glimpse was provided into the time course of cognitive processing in both experiments than the ERP technique. We found that microstate 3 corresponding to N170, ended later in the fear condition, which can be considered as a fear capture effect. In microstate 5, which corresponds to LPC, we observed that it lasted longer under fear conditions. microstate 5 represented the high-level cognitive stage of facial meaning estimation and was associated with task requirements involving evaluation and memory coding (Frühholz, Jellinghaus, & Herrmann, 2011). The higher-level cortex thus appeared to allocate more top-down cognitive resources to the recognition of the fear emotion. Overall, the attentional state could not prevent N170 and LPC from being modulated by emotions, regardless of whether the subjects needed to respond to facial information or not. In the audiovisual integration of meaningless sounds in different attentional states, the fear emotion regulation effect on the ERP components was similar to that of the visual stimuli presented alone.

### 2.3. The emotional audiovisual model

The neurobiological model proposed by Brück et al. (2011) aimed at explaining the neural processing mechanism of multimodal affective information processing. The model was divided into three main stages: information extraction, audiovisual integration/recognition, and making a judgment (Brück et al., 2011; Wildgruber et al., 2009). From an EEG perspective, our P1 and microstate 1 results in the two experiments are consistent with the first phase of the model, representing visual perception. Further, the P1 findings on the interaction between emotion and sound indicate that such integration or exchange of information may occur in the first stage. Falchier, Clavagnier, Barone, and Kennedy (2002) proposed that horizontal connections exist among the primary sensory areas and that multisensory interaction can occur at any stage. At the face structural coding phase, N170 exhibited the under-additivity property in the audiovisual condition, meaning that meaningless sounds affected this phase, in the line with the multisensory integration hypothesis of the second stage of the model. At the final stage, the results of the two experiments were significantly different. Experiment 1 required participants to judge facial emotion, causing the microstate 5 to appear, which is a topographic map showing parietal cortex activity, which indicates emotion estimation. While Experiment 2 required participants to judge the position of a bar, we speculate that the cognitive strategy adopted by the participants was to continuously gaze and scan the physical information of the face. This served to keep more positional information in working memory, and then to compare the remembered position with the bar that appeared later. Therefore, in Experiment 2, we observed activation of microstate 1 instead of microstate 5, due to activation of the visual cortex in the occipital lobe (Fig. 6). The difference between the two experiments was also reflected in the LPC results. In the explicit task, participants were required to extract emotion-related memories, resulting in a stronger LPC amplitude at the electrode sites in the parietal lobe (Friedman & Johnson, 2000). The results of the ERP and microstate analysis in the final stage fully supported the neurobiological model, which posits that the higher-order cortex is responsible for the cognitive evaluation of explicit emotional memory or meaning (Ethofer et al., 2009). In implicit emotion processing, the amygdala and nucleus accumbens are more sensitive to the marginal structure, and the processed information does not pass through the dorsolateral prefrontal cortex. Overall, the processing course revealed by the two experiments also supported the hypothesis of parallel integration. From the perspective of attention, MSI could occur in either the early or late stages according to the parallel integration model (Calvert & Thesen, 2004). Early integration would be mostly automated, bottom-up processing, while late integration would be regulated top-down and require conscious awareness.

Combining the two models, we propose three stages of audiovisual integration for emotion decoding. First, in the extraction stage, the audio and visual cortices encode external information separately and from the bottom-up way without being affected by the attention state, and information interchange between them is possible. At this stage, the information exchanged by the two modalities can provide a rough, simple, and fast perception of emotion, and sound changes (information consistency, elevated tone, etc.) can inhibit or facilitate processing on faces. Secondly, at the face-structure processing stage, emotion-related subcortical nuclei such as the amygdala and nucleus accumbens utilize the information from the integration cortex to refine the facial structure and contour information. The facial meaning assessment stage that follows is divided into two parts. The first mainly classifies and evaluates the face valence and degree of arousal in greater detail. In the second, the high-level cortex allocates cognitive resources top-down according to task requirements and makes different processing choices for different current stimuli (Cecere et al., 2017).

## 6. Conclusion

Overall, our experiments provide evidence for the role of meaningless sounds in promoting emotional face processing. Disruption of sound destroys information continuity in the auditory modality, which improves the participant's ability to allocate attention at the early stage of processing, resulting in bottom-up enhancement of attention distribution, independent of task requirements or any top-down effects requiring conscious awareness. In contrast, we found that in the later, cognitive stages of processing, participants would adjust the processing strategy top-down according to the needs of the task. Also, the present study explored the fear emotions effect on various ERP components in different attentional states, with results similar to those of visual stimuli alone. Finally, we propose three main stages of processing meaningless sound presented in conjunction with emotion-expressing faces, which provides a theoretical framework for future research. To sum up, the study of emotional audiovisual integration using interrupted sound can help us to better understand the promoting effect of meaningless sound on emotional face processing and the neurological mechanisms involved.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopsycho.2021.108246.

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